Jev-Backed Function Calling: Agentic In-Context Learning with Cumulative Decision Feedback

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Abstract

Function-calling models must choose a function and input values that match the user's request. Correct format alone does not ensure correct choices. We present Jev-Backed Function Calling (**JBFC**), an inference-time method that uses TypeSafe's System One (Decision) model *Jev* to check an LLM's proposed call. *Jev* checks function choice and, when a call is present, its arguments with binary questions. If a required check fails, we add the proposal and judgments to a growing history placed before the original user request. The LLM uses this feedback to revise both the function and arguments. Acceptance requires all applicable binary checks and the schema check to pass on the same proposal. The repair limit is fixed at three. Our code compares seven methods on a 3,051-example Berkeley Function Calling Leaderboard (BFCL) subset, measuring call correctness, function choice, argument correctness, time, and cost. In one run with `gpt-5-nano`, **JBFC** raises call accuracy over direct calling from 63.8% to 75.3%, mostly by avoiding unnecessary calls, at about five times the cost. It has the highest call accuracy of the seven methods, and keeping all earlier feedback adds 1.6 points over showing only the latest rejection.

1 Introduction

To carry out a user's request correctly, a function-calling system must choose the right function and provide the right argument values. A language model, called the generator, makes these choices. However, a call can have the required format and still use the wrong function or values. For example, a travel assistant may choose the wrong booking function or swap the departure and destination. A JSON schema check verifies required inputs and their types, but cannot guarantee that the call matches the user's request.

Tool-use training and evaluation have improved function-calling systems [2, 3, 1]. Even so, the generated function and arguments can still be wrong. We study whether checks from a separate decision model can guide the LLM to improve its next proposal before execution.

JBFC uses *Jev*, TypeSafe's System One model, to answer binary questions about an LLM's function choice and arguments [12]. When a check fails, we keep the proposed call and its judgments and place all earlier records before the original user request. The LLM then proposes a new call using this accumulated feedback. It can change both the function and argument values. **JBFC** accepts a proposal only when all applicable binary checks and the schema check pass on that proposal.

Earlier work already uses checks and feedback to fix calls. ToolVerifier checks tool choice and parameters [9]; ToolCritic gives feedback on tool-use errors [10]. ReAct uses reasoning, actions, and observations to guide later steps [16]. **JBFC** also carries feedback forward in the prompt, but its feedback comes from checking proposed calls before execution. Our contribution is this specific use of two *Jev* judgments and accumulated prompts, together with an evaluation of its effects and costs.

We compare **JBFC** with direct calling, schema repair, self-revision, single-check versions, and a latest-feedback version. A fixed repair limit of three controls the comparison. We measure errors fixed or introduced by repair, and keep call correctness separate from acceptance, time, and cost.

2 Related Work

Tool use and benchmarks. Gorilla studies API-call generation [2]; ToolLLM studies tool-use training and evaluation [3]. BFCL tests function calling across many settings [1]. We use its single-turn Python categories to study decisions made before a function runs. API-Bank offers dialogues with executable API calls [4]. StableToolBench uses caching and simulation to handle changing APIs [5]. These suggest broader tests beyond our BFCL subset.

Checks and feedback. Self-Refine uses self-generated feedback [6], Reflexion uses written feedback [7], and CRITIC uses feedback from tools [8]. Self-correction can turn correct answers into wrong ones [11], so we measure both fixes and new errors. ToolVerifier compares alternatives to check tool choice and parameters separately [9]. ToolCritic identifies tool-use errors and returns feedback [10]. **JBFC** checks complete proposals with a fixed external model through an API and limits repairs.

Accumulated prompts. ReAct combines reasoning with actions and observations from an environment [16]. Our method uses a similar idea of carrying earlier feedback into the next prompt. Each record contains a proposed function call and verification results, rather than an execution result. The LLM receives all records before the unchanged user request. We do not require a generated reasoning trace or run functions during repair. Many-shot in-context learning often improves as prompts grow to hundreds or thousands of examples, and model-generated rationales kept only when their answers are correct can stand in for human-written ones [17]. Our records are likewise model-generated examples, labeled by *Jev* instead of reference answers.

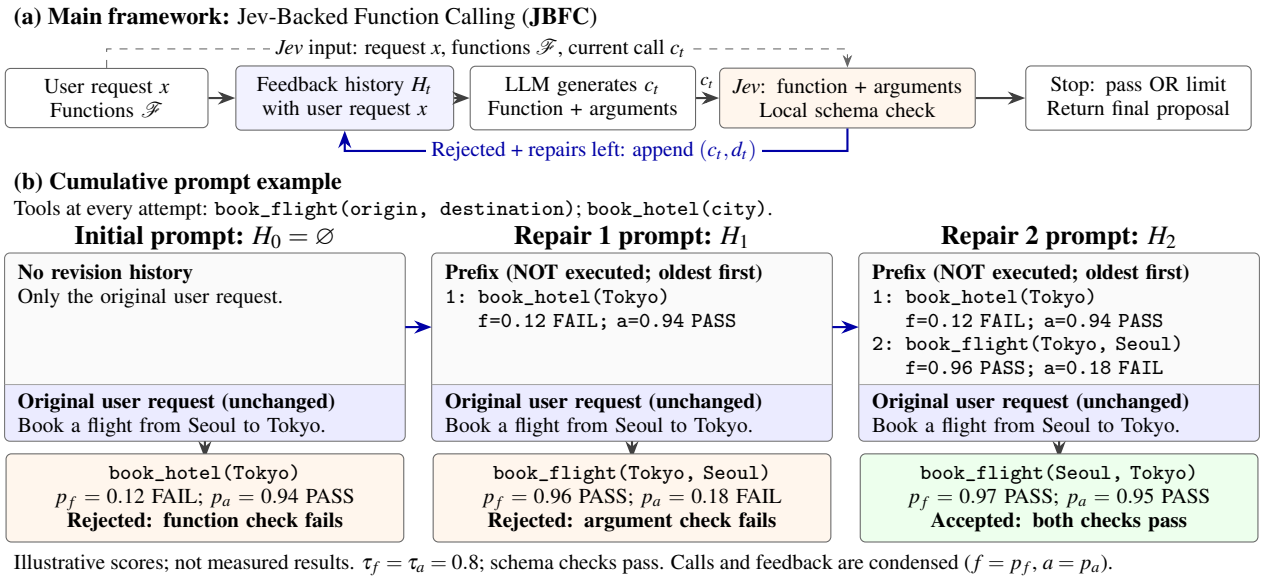


Figure 1: **JBFC** and cumulative prompt feedback. Top: the LLM reads earlier calls and judgments (H_t), then the original request (x), and generates a call (c_t) with a function and arguments. Failed checks add a record for another attempt. Bottom: growing history precedes the unchanged request; new calls appear below. Argument checks concern the proposed function. No function runs during repair.

***Jev* as a judge.** Li et al.'s *JEV-as-a-Judge* [15] checks finished outputs with probabilities for defined verdicts. If a verdict is uncertain, an independent, stronger judge decides. With fixed rules for when to use the stronger judge, it lowers judging cost while keeping similar accuracy on some tasks. It has limits on difficult reasoning, misleading answer styles, and writing judged without reference information. The study does not test tool or action selection inside agents. **JBFC** checks proposed function calls and returns failed checks to the same generator for revision. We test the specific design of **JBFC**.

3 Method

3.1 Propose and Check a Call

Let x be the user request, $\mathscr{F}$ the available functions, and G the generating LLM. A proposal is $c_t = (f_t, a_t)$, with function f_t and arguments a_t, or $c_t = \varnothing$ for no call. We study single-call-or-no-call decisions.[1] Local code checks readability, function availability, required arguments, and types. Write $S(c_t) = 1$ when these checks pass.

Binary verification: JBFC. For each readable proposal, one *Jev* request checks the function choice and, when a call is present, its arguments:

$$c_t = G(H_t, x, \mathscr{F}), \tag{1}$$

$$p_f^{(t)} = J_f(x, \mathscr{F}, c_t), \qquad p_a^{(t)} = J_a(x, \mathscr{F}, c_t). \tag{2}$$

Here H_t contains earlier proposals and feedback and is initially empty. The function `Noul` question asks whether the function choice, including no call, fits the request. The argument question checks required values, their support in the request, and their parameter roles. For no call, it is skipped and $p_a^{(t)}$ is undefined. A selected function with missing arguments still receives both checks. Both must pass for a call.

3.2 Accept or Repair a Call

Acceptance requires the schema check and all required *Jev* checks to pass on the same proposal. The acceptance rule is

$$A_t = \begin{cases} S(c_t)\,\mathbf{1}[p_f^{(t)} \geq \tau_f], & c_t = \varnothing, \\ S(c_t)\,\mathbf{1}[p_f^{(t)} \geq \tau_f]\,\mathbf{1}[p_a^{(t)} \geq \tau_a], & \text{otherwise.} \end{cases} \tag{3}$$

`Noul` returns the probability of yes, with no separate confidence field [13, 14]. We start with $\tau_f = \tau_a = 0.8$, untuned defaults chosen for this experiment.

Feedback before the user request. H_t stores earlier records. The current record (c_t, d_t) pairs proposal c_t with feedback d_t (probabilities, pass/fail flags, thresholds, and local errors). No-call skips the argument check. After rejection, if repairs remain, $\|$ adds $[(c_t, d_t)]$, a list containing this one record, to the end of H_t:

$$H_{t+1} = H_t \,\|\, [(c_t, d_t)]. \tag{4}$$

[1] We treat single-turn decisions as the basic unit of function calling. Parallel calling combines multiple such decisions, while multi-turn interactions repeat them with updated context. We therefore evaluate the verification and repair loop on this basic unit and exclude parallel and multi-turn settings from this study.

Algorithm 1 JBFC: verification and prompt repair

Require: Request x, functions $\mathscr{F}$, generator G, *Jev* J, repair limit r, thresholds τ_f, τ_a

$H \leftarrow [\,]$
for $t = 0, \ldots, r$ **do**
  $c \leftarrow G(\text{Prefix}(H, x), \mathscr{F})$
  $(s, e, \text{readable}) \leftarrow \text{SchemaCheck}(c, \mathscr{F})$
  $p_f, p_a \leftarrow \text{null}, \text{null}$
  if readable **then**
    $Q \leftarrow \{\text{function}\}$
    if $c \neq \varnothing$ **then** $Q \leftarrow Q \cup \{\text{arguments}\}$
    $(p_f, p_a) \leftarrow J(x, \mathscr{F}, c, Q)$
  end if
  $A \leftarrow \text{Accept}(s, c, p_f, p_a, \tau_f, \tau_a)$ ▷ Eq. 3
  $d \leftarrow \text{Feedback}(e, p_f, p_a, \tau_f, \tau_a, c)$
  if $A = 1$ **or** $t = r$ **then break**
  $H \leftarrow H \,\|\, [(c, d)]$
end for
return c

The LLM reads this history before the unchanged request and can change both function and arguments. Schemas remain available; system instructions stay separate.

Algorithm 1 rechecks up to $r+1$ proposals: the first plus repairs ($r = 3$ by default and in experiments). Its `Prefix` function places JSON records, oldest first, before unchanged original messages. Empty history adds no prefix. Records include round numbers and mark proposals as not executed. Here $s = S(c)$ and e lists local errors. Readable proposals receive *Jev* checks even if their schema fails; unreadable ones receive local feedback. *Jev* never receives history. Unasked p_a stays null. Null required probabilities cannot pass the gate; invalid API responses stop the run. Reference answers stay out of model requests. API errors stop the run after transport retries, separate from repairs.

3.3 Final Proposal

When a proposal passes all applicable checks, we return it. If the repair limit is reached first, we return the last proposal even if it fails a check (`execute_last`). We score the final proposal against the benchmark answers without running the function.

4 Experiments

4.1 Data and Evaluation

We prepare 3,051 BFCL examples from seven categories at repository revision `6ea57973c7a6`[2]. Each run uses all examples once, with *Jev* thresholds fixed at 0.8. The repair limit $r = 3$ was chosen on the first run, after Figure 2: beyond it, each extra point costs about four times as much.

The original Python abstract syntax tree (AST) checker compares function names and arguments with allowed answers, not execution success; irrelevance categories require no call. Live relevance checks only call presence, so the main score, a micro-average, covers 3,035 examples. BFCL's "multiple" categories offer several functions but no parallel calls; we exclude parallel, multi-turn, web-search, memory, and execution categories.

[2]Full revision: `6ea57973c7a6097fd7c5915698c54c17c5b1b6c8`. The saved setup lists source URLs, hashes, and format changes.

		`gpt-5-nano`		`gpt-5.4-mini`	
Category	Examples	Direct	**JBFC**	Direct	**JBFC**
Simple Python	400	90.8	91.0	90.5	**91.2**
Multiple	200	91.0	91.5	**93.3**	**93.3**
Irrelevance	240	27.2	81.7	**87.1**	81.1
Live simple	258	74.3	75.1	77.1	**79.1**
Live multiple	1053	67.6	70.4	75.2	**76.4**
Live irrelevance	884	48.8	67.7	**80.7**	74.4
Live relevance[†]	16	**89.6**	87.5	79.2	85.4
Total	3051	64.1±0.3	75.0±0.2	**81.1**±0.2	79.5±0.2
USD/1k		0.056	0.272	0.703	1.700

Table 1: Accuracy (%) on all BFCL examples, mean of three runs, $r = 3$ (Direct makes no repairs). Bold: highest per row. Total: mean ± s.d. across runs. USD/1k: mean cost per 1,000 requests. [†]Call presence only; not in the total.

4.2 Models and Methods

The generator is the Azure OpenAI deployment `gpt-5-nano` (`gpt-5-nano-2025-08-07`), with reasoning effort `none` and at most 4,096 output tokens; *Jev* is pinned to `jev-1.13.0`. We use Chat Completions and System One through an APIM gateway (System One directly once the gateway stopped serving it) and convert BFCL schemas to JSON Schema with API-safe function names. Generations cut off at the output limit or blocked by the provider's filter count as wrong. Direct and **JBFC** also run on the larger, costlier deployment `gpt-5.4-mini` (snapshot 2026-03-17) with the same settings. Table 1 averages three runs per generator, each with new initial proposals.

We compare seven methods, sharing one initial proposal per example. *Direct* uses the initial proposal; *Schema-only* repairs schema errors. *Self-refine* uses generic self-review in all r repairs, not the full Self-Refine method [6]. **JBFC-F** and **JBFC-A** obtain applicable `Noul` judgments but expose and require only the function or argument verdict, plus schema checks. **JBFC-A** accepts no call on schema checks alone. **JBFC** requires all applicable checks. **JBFC-L** shares its settings and initial proposal but shows only the latest rejected proposal and feedback. Other repair methods show all earlier records before the request. With $r = 3$, repair methods make at most four proposals, and the **JBFC** and **JBFC-L** prompts differ from the second repair on. Neither model sees r and nothing runs before the end, so a larger limit only continues trajectories: the first run ran all methods with $r = 2$ and continued them (**JBFC** to $r = 10$ for Figure 2).

4.3 Measures, Cost, and Uncertainty

We report call, function, and argument accuracy (the last given the correct function), fixes and new errors from initial to final proposals, acceptance, and repairs. We compare repairs with Direct and **JBFC** with **JBFC-L** on matching examples and initial proposals, bootstrapping example IDs 2,000 times (seed 1729) for 95% intervals after averaging each example over runs.

Method	Call	Func.	Args.	Sec.	USD/1k
Direct	63.8	74.8	81.2	1.30	0.056
Schema repair	65.8	76.4	81.8	1.36	0.059
Self-refine	70.0	80.3	82.2	5.47	0.265
JBFC-F	73.6	84.6	81.6	2.99	0.223
JBFC-A	73.7	83.5	83.1	2.66	0.199
JBFC-L	73.6	84.0	82.7	3.47	0.262
JBFC	**75.3**	**85.0**	**83.4**	3.49	0.273

Table 2: Methods on all BFCL examples (one run, $r = 3$, `gpt-5-nano`). Bold: highest accuracy per column. Call and function accuracy (%) exclude live relevance; argument accuracy (%) uses calls with the correct function.

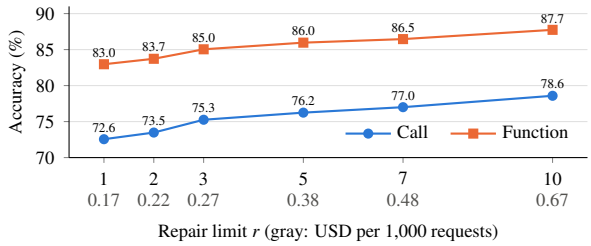


Figure 2: **JBFC** accuracy (%) by repair limit r: one run continued to $r = 10$ and cut at each limit ($r = 3$ is Table 2); gray: USD per 1,000 requests. Repairs 8–10 call *Jev* directly.

Costs count all generation and verification work, including the growing history and the shared initial call; times are mean API seconds without retry waits. Prices in USD per million uncached/cached/output tokens are 0.05/0.005/0.40 for `gpt-5-nano` and 0.75/0.075/4.50 for `gpt-5.4-mini`; *Jev* input costs $0.042 (output free), without gateway fees.

4.4 Results

In Table 1, **JBFC** raises mean accuracy from 64.1% to 75.0% (paired difference 10.9 points, 95% interval 9.8–12.1; per run 11.5, 10.6, 10.6). Per run, it fixes 374–399 wrong initial proposals, breaks 50–55, and averages 1.07–1.08 repairs; 31% of examples reach the limit without passing, and the 69% that pass are 91% correct. Other numbers are from the first run unless stated.

A larger generator. Without *Jev*, `gpt-5.4-mini` reaches 81.1%, 6.1 points above **JBFC** with `gpt-5-nano` (95% interval 4.9–7.4), at 2.6 times the cost and a third of the time. **JBFC** lowers its accuracy in every run, by 1.6 points on average (95% interval 0.9–2.4): repairs fix more calls than they break where a call is needed (55–60 vs. 36–39), but *Jev* separates its remaining unnecessary calls poorly (median function probability 0.72–0.83 vs. 0.85–0.94 for correct no-calls), and its rejections turn 102–110 correct no-calls into calls per run.

Comparison of methods. In Table 2, every repair method beats Direct (paired 95% intervals above zero): schema repair by 2.0 points, self-refine 6.3, **JBFC-F** 9.9, **JBFC-A** 10.0, and **JBFC-L** 9.9. Self-refine makes four proposals per request yet trails the *Jev*-based methods (1.7 to 2.1).

Why no-call cases gain most. Each Irrelevance example offers one unrelated function; Direct calls it in 179 of 240 cases. *Jev*'s function question separates them from the 61 correct no-calls (median probability 0.09 vs. 0.95); **JBFC** rejects 166, and the generator withholds 137. In live irrelevance (real requests, 2.3 functions), *Jev* separates less sharply (0.53 vs. 0.87); the generator drops 198 of 432 rejected calls, and 25 correct no-calls become calls. Where reference calls exist, **JBFC** rejects 334 of 463 wrong initial calls but fixes only 64 and breaks 25: a binary verdict says a call is wrong, not which call is right, but an unnecessary call has an obvious repair.

More repairs. In Figure 2, accuracy reaches 78.6% at $r = 10$, at 2.1 times the time and 2.4 times the cost of $r = 3$; of the 148 examples that turn correct (47 turn wrong), 128 are irrelevance cases that end without a call; only 33 pass.

Records as in-context examples. Each record pairs a generated call with *Jev*'s verdict, so the history acts as labeled in-context examples, made without reference answers in 0.30 s per label (the generator takes 1.38 s) at about $0.06 per 1,000. As in many-shot learning [17], more examples help: with the same attempts, all records beat the latest one by 1.6 points (**JBFC** minus **JBFC-L**, 95% interval 0.8–2.5), and accuracy rises with the limit, which also adds attempts. Unlike many-shot prompts, these are failed attempts at the same request; help for other requests is untested.

4.5 Study Limits

The verifier can accept a wrong call or share the generator's mistake, and repairs or growing feedback can introduce or reinforce errors; higher acceptance alone does not show improvement. We use two generators, one verifier, and mostly English single-turn requests; Table 2 and Figure 2 come from one run, and public examples may appear in training data. Ten documented annotation errors stay (six block schema-checked methods). Thresholds and effort `none` are untuned; r was chosen on the first run, **JBFC**'s best of three (75.3% vs. 74.9%), and one common limit may favor some methods.

5 Conclusion

Feeding back *Jev*'s judgments as a history, **JBFC** with `gpt-5-nano` and three repairs raised mean accuracy from 64.1% to 75.0% at about five times the cost, mostly by withholding unnecessary calls; in one run, the full history beat the latest feedback by 1.6 points. With `gpt-5.4-mini`, **JBFC** lowered accuracy in every run (81.1% to 79.5% on average): rejections of correct no-calls outweighed its fixes.

Acknowledgements

This work was created as part of the No-More-Slops group activities at Pseudo Lab.

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